

Riemann Sums, Accumulation, and the Fundamental Theorem

Answer Key

AP Calculus AB/BC Integration

Quick Answers

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|----------------|---------------------|
| 1. $x^4 - x^2$ | 6. $2x^2 + \sin(x)$ |
| 2. 80 | 7. -4 |
| 3. 26 | 8. 18, 9 |
| 4. 12 | 9. 5.75 |
| 5. 90 | 10. 90, 100 |
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Common wrong answers

2. If they got 100: used the right endpoints instead of the left ones
5. If they got 55: forgot to double the interior values and added the five readings once each
7. If they got 4: subtracted in the wrong order, computing $F(0) - F(2)$ instead of $F(2) - F(0)$
9. If they got 3.75: used the left endpoints instead of the right ones
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1. $\int (4x^3 - 2x) dx.$

Step 1. Antidifferentiation reverses the power rule term by term, so raise each exponent by one and divide by the new exponent rather than looking for a single pattern.

Step 2. $4x^3 \rightarrow \frac{4x^4}{4} = x^4$ and $-2x \rightarrow -\frac{2x^2}{2} = -x^2$.

$x^4 - x^2 + C$

Check by differentiating: $\frac{d}{dx}(x^4 - x^2) = 4x^3 - 2x$, the original integrand.

2. Left Riemann sum for the velocity table.

Step 1. A left sum uses the value at the *start* of each subinterval, so the reading at $t = 8$ is never used; four subintervals across $[0, 8]$ give $\Delta t = 2$.

Step 2. The left-endpoint velocities are 5, 9, 12 and 14, so the sum is $2(5 + 9 + 12 + 14) = 2(40) = 80$.

80

Interpretation: about 80 metres, treating the velocity as constant over each 2-second block.

The trap: using the right endpoints gives $2(9 + 12 + 14 + 15) = 100$; since v is increasing, that is the larger estimate and the true distance lies between the two.

3. $\int_1^3 3x^2 dx.$

Step 1. The integrand has an elementary antiderivative, so FTC part 2 applies: find F , then subtract.

Step 2. $F(x) = x^3$, and $F(3) - F(1) = 27 - 1 = 26$.

26

Note: the constant of integration cancels in the subtraction, which is why an antiderivative — any one of them — is enough.

4. $F(x) = \int_0^x (2t + 1) dt$, find $F(3)$.

Step 1. F is an accumulation function: the upper limit is the input, so evaluating F at 3 just means integrating from 0 to 3.

Step 2. An antiderivative of $2t + 1$ is $t^2 + t$, so $F(3) = (9 + 3) - (0 + 0) = 12$. 12

Part (b): $F(3)$ is the signed area under $y = 2t + 1$ from $t = 0$ to $t = 3$ — a trapezoid with parallel sides 1 and 7 and width 3, whose area is indeed 12.

5. Trapezoidal estimate for the velocity table.

Step 1. A trapezoidal sum averages consecutive readings, which is the same as weighting every interior reading twice; write the coefficient pattern before substituting.

Step 2. $\frac{\Delta t}{2} [v_0 + 2v_1 + 2v_2 + 2v_3 + v_4] = \frac{2}{2} [5 + 2(9) + 2(12) + 2(14) + 15]$.

Step 3. $= 1 [5 + 18 + 24 + 28 + 15] = 90$. 90

Why the interior values double: each interior reading is the right end of one trapezoid and the left end of the next, so it appears in two terms. *The trap:* counting each reading once gives 55, which is far below even the left sum and should be rejected on sight.

6. $\int (\cos x + 4x) dx$.

Step 1. The integrand is a sum of two unrelated pieces, so antidifferentiate each separately rather than searching for a substitution.

Step 2. $\cos x \rightarrow \sin x$ and $4x \rightarrow \frac{4x^2}{2} = 2x^2$. sin x + 2x² + C

Check: differentiating gives $\cos x + 4x$.

7. $\int_0^2 (x^3 - 4x) dx$.

Step 1. FTC part 2 again; the order of subtraction is fixed as $F(\text{upper}) - F(\text{lower})$, and reversing it flips the sign of the whole answer.

Step 2. $F(x) = \frac{x^4}{4} - 2x^2$, so $F(2) = 4 - 8 = -4$ and $F(0) = 0$.

Step 3. $F(2) - F(0) = -4 - 0 = -4$. -4

Part (b): on $(0, 2)$ the curve $y = x^3 - 4x$ lies *below* the x -axis, so the definite integral counts that region negatively. The definite integral measures signed area; total geometric area would be $|-4| = 4$, and the two answer different questions. *The trap:* computing $F(0) - F(2)$ gives $+4$ — the right magnitude with the wrong sign.

8. $g(x) = \int_1^x f(t) dt$ with $f(t) = 3t^2 - 6t$.

Step 1. Part (a) asks for a value of the accumulation function, so integrate; part (b) asks for its derivative, so do *not* integrate — FTC part 1 gives the derivative directly.

Step 2. An antiderivative of f is $t^3 - 3t^2$, so $g(4) = (64 - 48) - (1 - 3) = 16 + 2 = 18$. 18

Step 3. By FTC part 1, $g'(x) = f(x)$, so $g'(3) = 3(3)^2 - 6(3) = 27 - 18 = 9$. 9

Part (c): $g'(3) = 9 > 0$, so g is increasing at $x = 3$. Equivalently, f is positive there, so the accumulation is still adding area.

9. Right Riemann sum for $f(x) = x^2 + 1$ on $[0, 2]$, $n = 4$.

Step 1. With $n = 4$ on an interval of length 2, $\Delta x = 0.5$, and a right sum uses the values at $x = 0.5, 1, 1.5, 2$ — the value at $x = 0$ is not used.

Step 2. Those four values of f are 1.25, 2, 3.25 and 5, so the sum is $0.5(1.25 + 2 + 3.25 + 5) = 0.5(11.5) = 5.75$. 5.75

Part (b): the exact value is $\frac{14}{3} \approx 4.67$, so 5.75 is an **overestimate**.

Part (c) — manual review. A correct justification does not compare the two numbers. On each subinterval f is increasing, so the right endpoint is where f is largest on that subinterval. Each rectangle therefore has height at least $f(x)$ for every x in its base, so each rectangle contains the region under the curve above its base, and the sum of the rectangles is at least the integral. The argument uses only that f is increasing, so it holds for any increasing function — and reverses for a decreasing one. *The trap:* a left sum here gives 3.75, an underestimate, for the mirror image of the same reason.

10. Challenge: tank filling at $r(t) = 4t + 3$ gal/min.

Step 1. A rate integrated over time gives accumulated change, so part (a) is a definite integral of r ; the starting 10 gallons is not part of that integral because it accumulated before $t = 0$.

Step 2. An antiderivative of $4t + 3$ is $2t^2 + 3t$, so $\int_0^6 (4t + 3) dt = (72 + 18) - 0 = 90$. 90

Step 3. Part (b): the amount at $t = 6$ is the initial amount plus the accumulated change, $10 + 90 = 100$. 100

Part (c) — manual review. FTC part 2 says $\int_0^6 A'(t) dt = A(6) - A(0)$. Since $A'(t) = r(t)$, the integral of the rate is exactly the *difference* $A(6) - A(0)$, not $A(6)$. Rearranged, $A(t) = A(0) + \int_0^t r(u) du$: the first term carries everything that was in the tank before the clock started, and the integral carries everything added since. A rate function knows nothing about initial conditions, which is why the constant has to be supplied from outside — the same reason an indefinite integral carries a $+C$.